Double-Angle Formulas and Trigonometric Equations

Double-angle formulas

1 Given that $\cos\theta = -\frac{3}{5}$ and θ lies in Quadrant II, find:

a $\sin 2\theta$ **b** $\cos 2\theta$ **c** $\tan 2\theta$

2 Use a double-angle formula to write $\cos^2 x$ in terms of $\cos 2x$, and state the corresponding formula for $\sin^2 x$.

Solving trigonometric equations

3 Solve $2\sin x - 1 = 0$ on the interval $[0, 2\pi)$.

4 Solve $2\cos^2 x - \cos x - 1 = 0$ on the interval $[0, 2\pi)$.

5 Solve $\sin 2x = \sin x$ on the interval $[0, 2\pi)$.

6 Solve $\tan^2 x = 3$ on the interval $[0, 2\pi)$.

Half-angle formulas

7 Use a half-angle formula to find the exact value of $\cos 15°$ (using $15° = \frac{30°}{2}$).

8 Given $\cos\theta = \frac{7}{25}$ with $0 < \theta < \frac{\pi}{2}$, find $\sin\frac{\theta}{2}$ exactly.

Equations requiring double-angle substitution

9 Solve $\cos 2x + \sin x = 0$ on the interval $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.

10 Solve $\sin 2x - \cos x = 0$ on the interval $[0, 2\pi)$.

Applications of double-angle formulas

11 A projectile's range is given by $R(\theta) = \frac{v^2 \sin 2\theta}{g}$. Using the double-angle formula, explain why the maximum range occurs at $\theta = 45°$.

12 Given $\sin\theta = \frac{1}{3}$ with θ in Quadrant I, find $\cos 2\theta$ and $\sin 2\theta$ exactly.

More equation practice

13 Solve $4\sin^2 x - 3 = 0$ on the interval $[0, 2\pi)$.

14 Solve $\cos 2x = \cos x$ on the interval $[0, 2\pi)$, using $\cos 2x = 2\cos^2 x - 1$.

Half-angle formulas for tangent

15 State the half-angle formula for $\tan\frac{\theta}{2}$ in terms of $\sin\theta$ and $\cos\theta$ (the form avoiding a square root), and use it to find $\tan 15°$.

Equations with multiple angle types

16 Solve $\sin 2x + \sin x = 0$ on $[0, 2\pi)$.

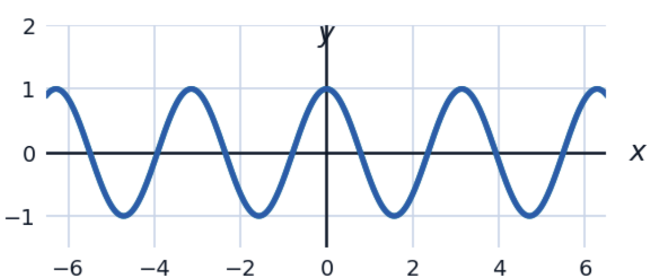
17 Solve $\cos 2x = 1 - 3\sin x$ on $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.

Triple-angle preview

18 Using $\sin(2x + x) = \sin 2x\cos x + \cos 2x\sin x$ together with the double-angle formulas, derive the triple-angle formula $\sin 3x = 3\sin x - 4\sin^3 x$.

Graphical verification of a double-angle identity

19 The functions $y = \cos 2x$ and $y = 1 - 2\sin^2 x$ are graphed together below. Explain what it means that the two curves appear identical, and state the identity this confirms.



Solving equations restricted to a smaller interval

20 Solve $2\cos 2x - 1 = 0$ for $x \in [0, \pi)$ only (note the restricted domain).